# Annotation Details and Criteria

Pacific low-frequency baleen whale networks

## OVERVIEW:

This document describes who annotated the training and evaluation data in this repository, how the annotation was done, and the criteria used for each call type, with example spectrograms. It supplements the README and the associated article. Questions about these details can be sent to Liz Ferguson (eferguson@oceanscienceanalytics.com).

## ANALYST TEAM:

All analysts are co-authors of this dataset.

- **Elizabeth (Liz) Ferguson:** experienced analyst (~20 years). Guided revisions, annotated earlier datasets and supported review.
- **Cory Ann Hom-Weaver:** experienced analyst (~16 years) with extensive knowledge of low-frequency baleen whale calls and instrumentation. Annotated mainly the CCES and DCLDE datasets and supported review.
- **Gabriela (Gabi) Alongi:** extensive experience with baleen whale calls. Guided sei whale call identification and adapted code to the needs of the multi-class network.
- **Julia Pelio:** analyst with ~4 years of acoustic data processing, experienced in annotating a range of baleen whale calls and some odontocete calls (e.g., killer whale burst pulses). Annotated most of the earlier fin whale datasets and all call types in this dataset.
- **Mattie Toll:** analyst with ~1.5 years of acoustic data processing and strong programming skills, particularly for iterative network training and evaluation. Annotated all call types in this dataset.

### Review process:

Each dataset was annotated by one analyst and reviewed by another. The team discussed the process throughout, reviewing selected examples together to agree on common practices. After each new dataset was added, a network was trained to check whether performance improved; if it did not, the team discussed how the bounding boxes might have affected the result. Each annotation dataset therefore went through several rounds of review and revision, until the team agreed on which calls were of sufficient quality to annotate and which to disregard.

# GENERAL ANNOTATION CRITERIA:

- Audio was decimated to a 400-Hz sample rate (0–200 Hz bandwidth) so that all datasets shared a common bandwidth in DeepAcoustics. CTBTO data were kept at their native 250-Hz sample rate (0–125 Hz).
- Annotation was iterative and took place over about two years. Fin whale 18-, 20- and 40-Hz pulses were annotated first; blue and sei whale calls were added once DeepAcoustics supported multi-class networks. Later effort focused on blue whale B and D calls, because fin whale annotations were already available from the earlier work.
- The networks are intended to detect both faint and loud calls regardless of background noise, so calls were annotated across a wide range of intensities and with as much natural variability as possible.
- Annotation remained conservative: questionable calls were not annotated. Faint calls were annotated where possible, but were disregarded if analysts could not agree on them or judged that they would confuse network training. Review ensured that annotations could be reproduced by an expert analyst trained to recognize these calls.
- Bounding boxes were drawn tightly around each call while keeping all of its elements inside the box, with a small buffer so the call stays inside the box under different FFT settings. Harmonics were included unless a call type below states otherwise.
- No noise samples were annotated. DeepAcoustics generates noise samples automatically from spectrogram areas outside the annotation boxes, so all potential calls in an annotated audio file need to be boxed.
- Training data are organized into folders by dataset; DeepAcoustics randomly sets aside a portion of the training annotations for validation during training. Evaluation (test) data are separate audio files, described in the README.
- All boxes were drawn in Raven Pro 1.6 with common spectrogram settings for the 400-Hz data (Figure 1).

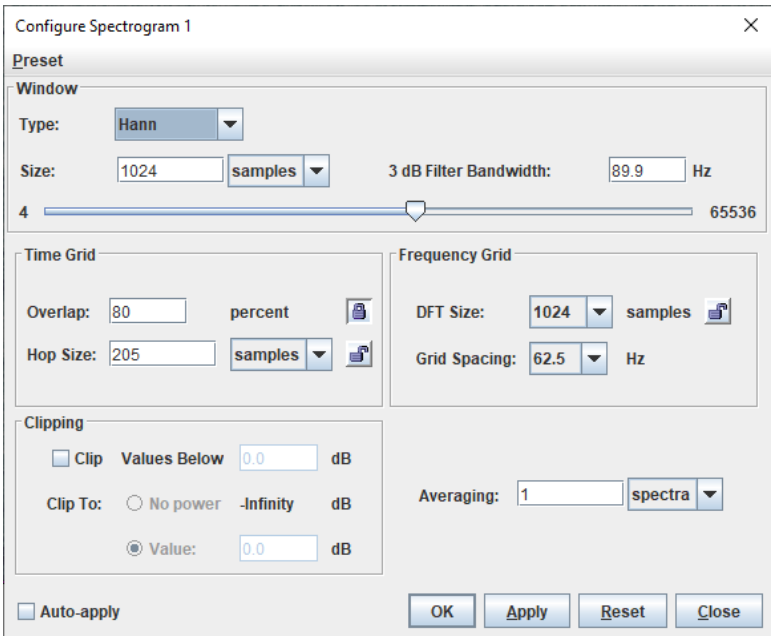



***Figure 1.*** *Raven Pro 1.6 spectrogram settings used for annotation: Hann window, 1024-sample window and DFT size, and 80% overlap (205-sample hop). The filter bandwidth and grid spacing shown depend on the sample rate of the file loaded when the screenshot was taken.*

# FIN WHALE ANNOTATION CRITERIA:

- Most fin whale calls encountered were 18- and 20-Hz pulses, with less frequent 40-Hz pulses. Occasional pulses centred near 30 Hz were too few for a separate category. In the literature, 18-, 20- and 30-Hz pulses are generally all described as 20-Hz pulses associated with song, while 40-Hz pulses have been suggested to be associated with feeding.
- Context mattered: pulses were annotated when they occurred as part of a series.
- Doublets and triplets of 18- and 20-Hz pulses were initially boxed together and labelled separately (e.g., "20Hz-d", "30Hz-t"). In the final annotations they were merged into the main call categories; the original labels remain in the Original_Annotation column.
- The waveform helped confirm the timing of faint pulses (Figure 2), and pulses too faint to annotate reliably were excluded (Figure 3).
- After successive rounds of training and discussion, tighter boxes around each pulse were found to improve results (Figure 4).
- 40-Hz pulses vary widely in minimum and maximum frequency and in bandwidth (Figure 7). They are distinguished from blue whale D calls mainly by their shorter duration (about 1 s; Figure 8).

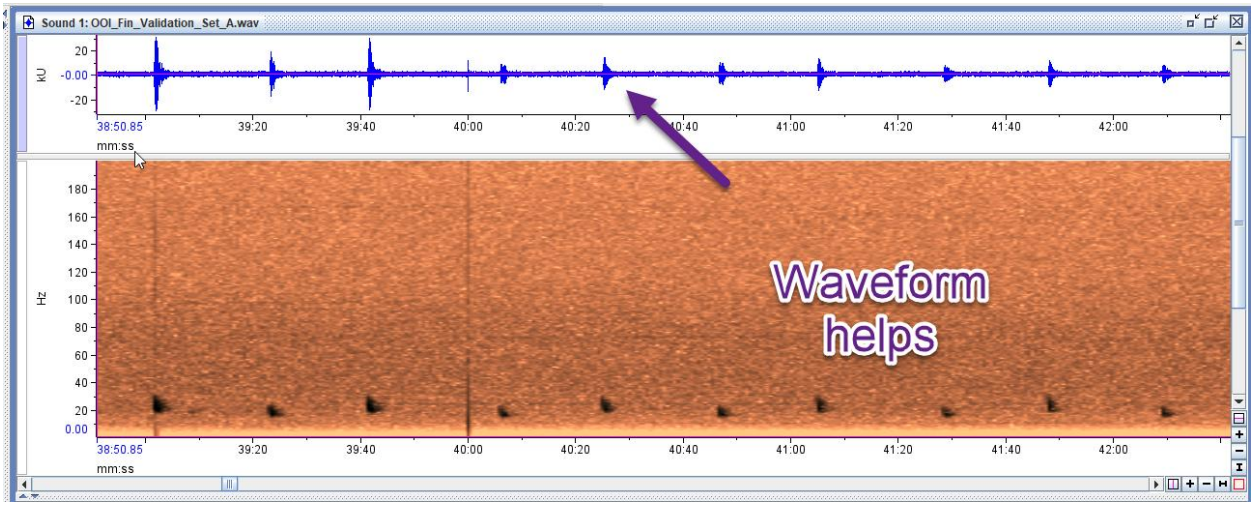



***Figure 2.*** *Waveform (top) and spectrogram (bottom) of a series of fin whale 20-Hz pulses in OOI data. Peaks in the waveform help confirm the timing of pulses that are faint in the spectrogram.*

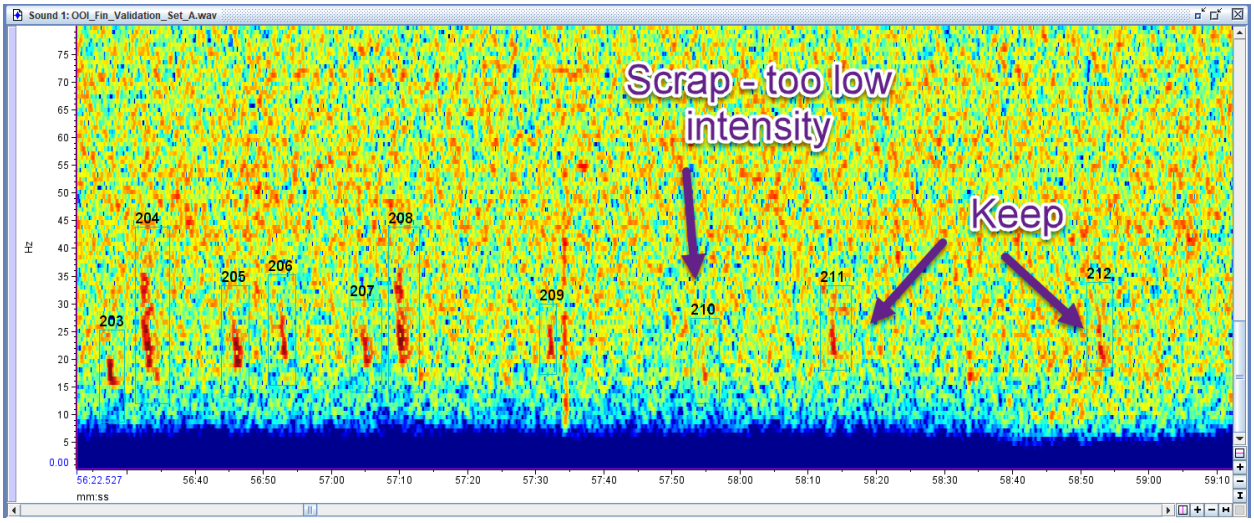



***Figure 3.*** *Fin whale 20-Hz pulse annotations in OOI data, showing an annotation decision: pulses with sufficient signal were kept (e.g., boxes 211 and 212), while a pulse too faint to annotate reliably was discarded (box 210).*

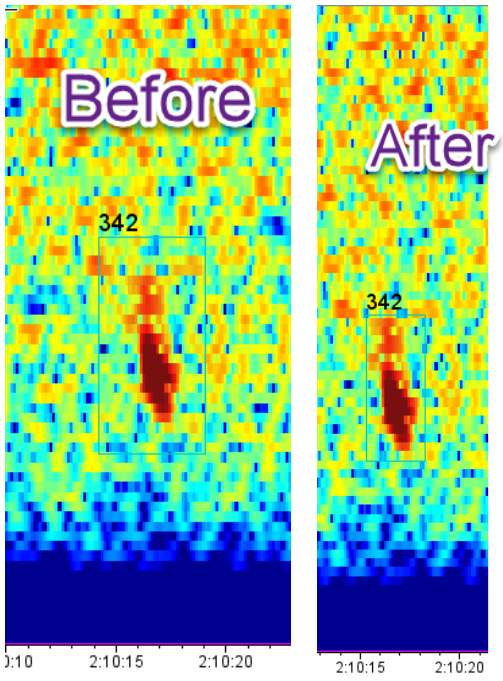



***Figure 4.*** *The same fin whale pulse (box 342) before (left) and after (right) the annotation was revised. Tighter boxes around each call improved network performance and were used in the final annotations.*

## Examples from the final fin whale annotations:

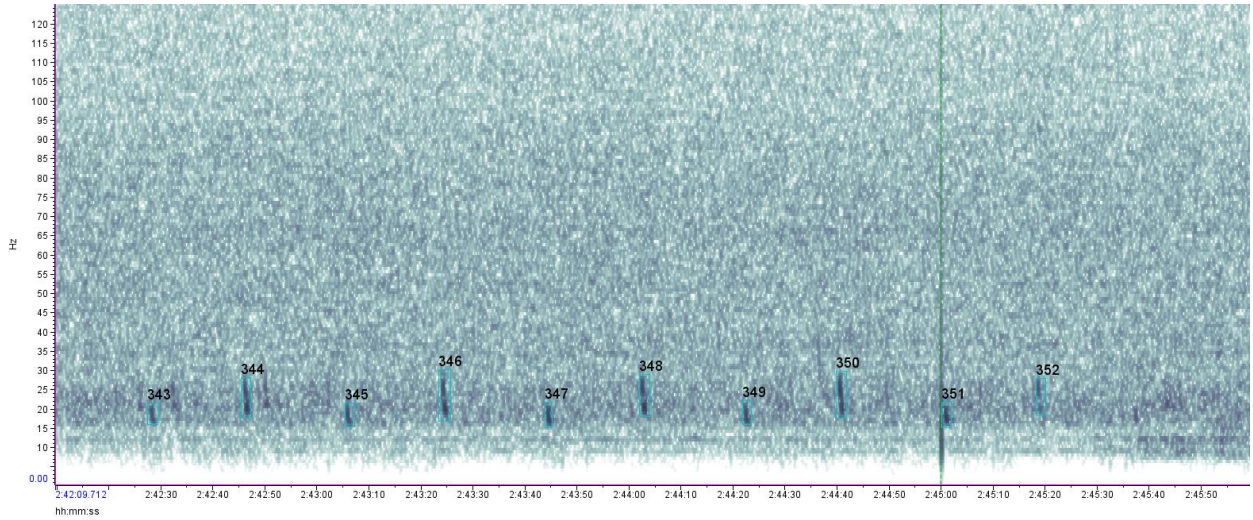



***Figure 5.*** *Final fin whale 18- and 20-Hz pulse annotations (boxes 343–352) in OOI data.*

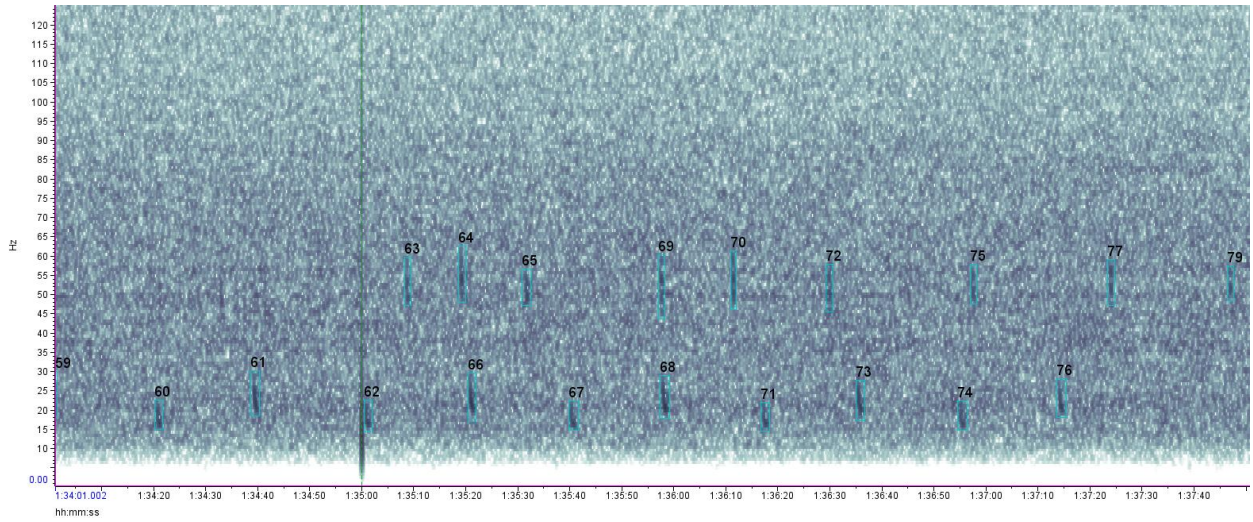



***Figure 6.*** *Final fin whale 20-Hz (lower boxes, ~15–30 Hz) and 40-Hz (upper boxes, ~45–60 Hz) pulse annotations in OOI data.*

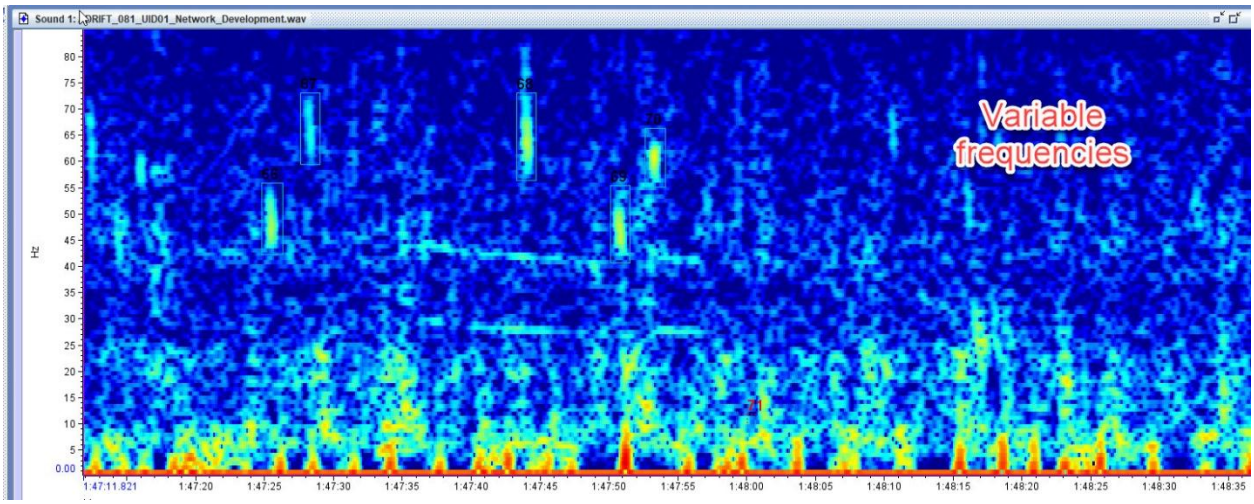



***Figure 7.*** *Fin whale 40-Hz pulse annotations, showing how widely the minimum and maximum frequency and bandwidth of these pulses vary.*

# BLUE WHALE ANNOTATION CRITERIA:

- **A calls:** only the component near 80 Hz, which is easy to distinguish in the data, was annotated.
- **B calls:** these were first annotated with the full set of harmonic bands, but this proved difficult for the networks. Detection improved when only the main harmonic, near 40–50 Hz, was annotated. B calls can last up to about 20 s and vary widely in intensity because they are detected over long distances.

- **Cut-off B calls:** B calls were sometimes cut off by the start of an audio file or masked by noise. In the CCES data these were first labelled "Cut Off B" (COB); in the final annotations they are labelled B, with the COB label kept in a separate column.
- **D calls:** these are highly variable downsweeps that occur singly or in sequences. They are distinguished from fin whale 40-Hz pulses mainly by their longer duration (more than about 2 s; Figure 8). Faint D calls were annotated when they could be clearly identified.

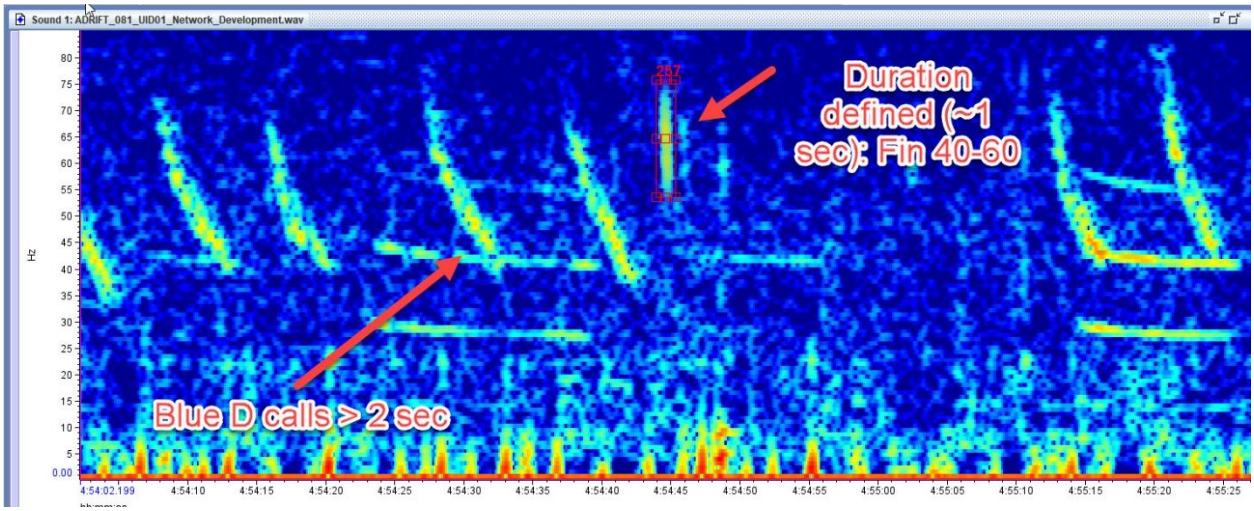



***Figure 8.*** *Blue whale D calls (downsweeps lasting more than about 2 s) alongside an annotated fin whale 40-Hz pulse (box 257; about 1 s, ~50–75 Hz). Call duration is the main feature separating these two call types.*

## Examples of blue whale A and B annotations:

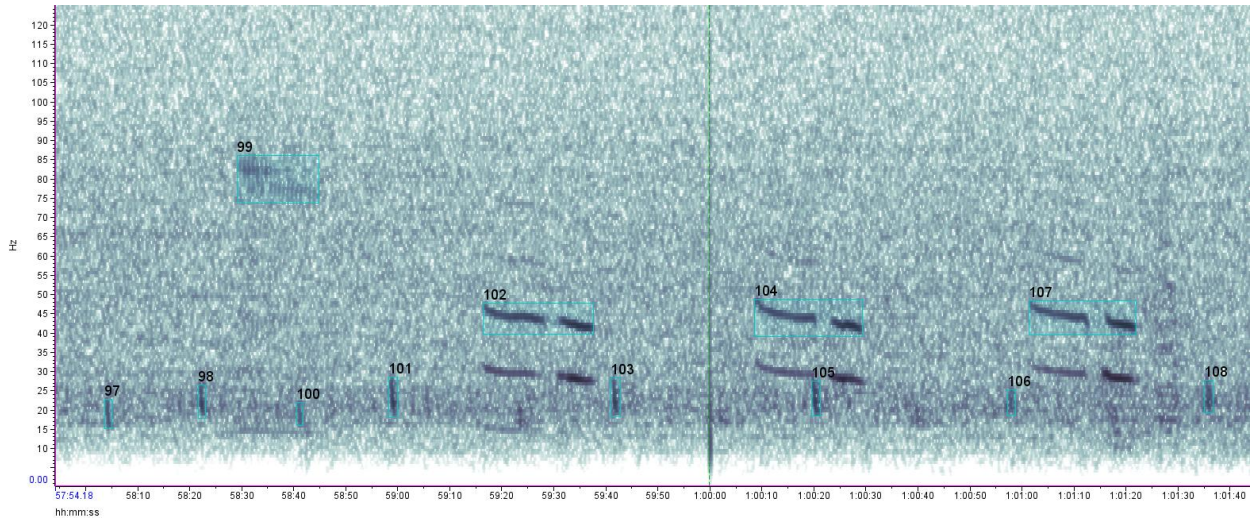



***Figure 9.*** *Blue whale A call (box 99, ~75–85 Hz) and B calls (boxes 102, 104 and 107; main harmonic ~40–50 Hz) annotated alongside fin whale 20-Hz pulses (lower boxes, ~15–25 Hz).*

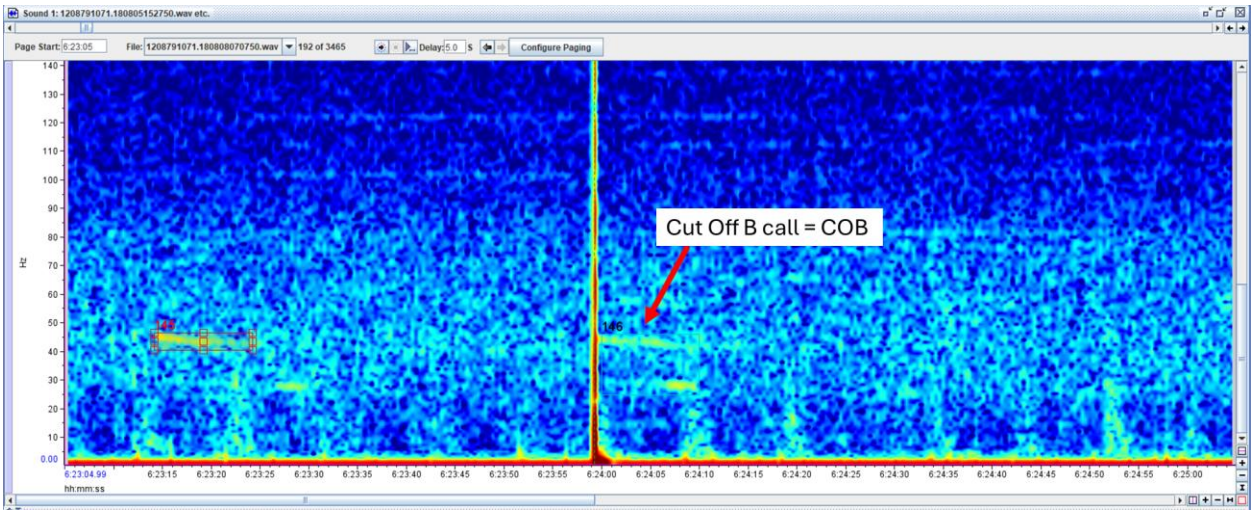



***Figure 10.*** *Blue whale B call cut off at the start of an audio file (box 146) in CCES data, next to a complete B call (box 145). Cut-off calls were first labelled "Cut Off B" (COB) and later relabelled B.*

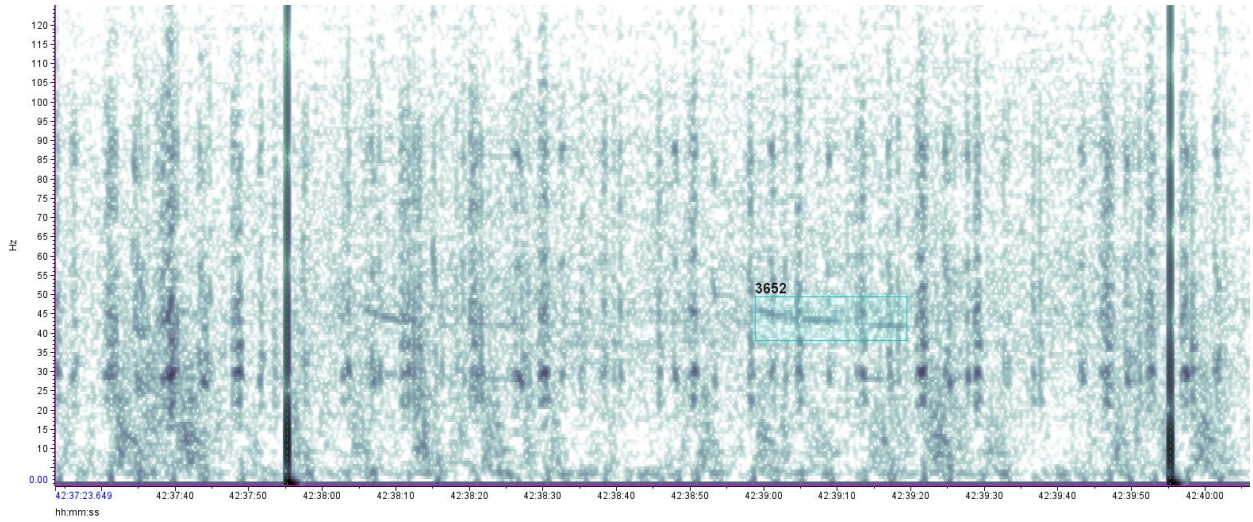



***Figure 11.*** *Annotation decision in noisy data: a blue whale B call was annotated (box 3652), while a possible B call just before it was left unannotated because it could not be confirmed.*

## Examples of blue whale D annotations:

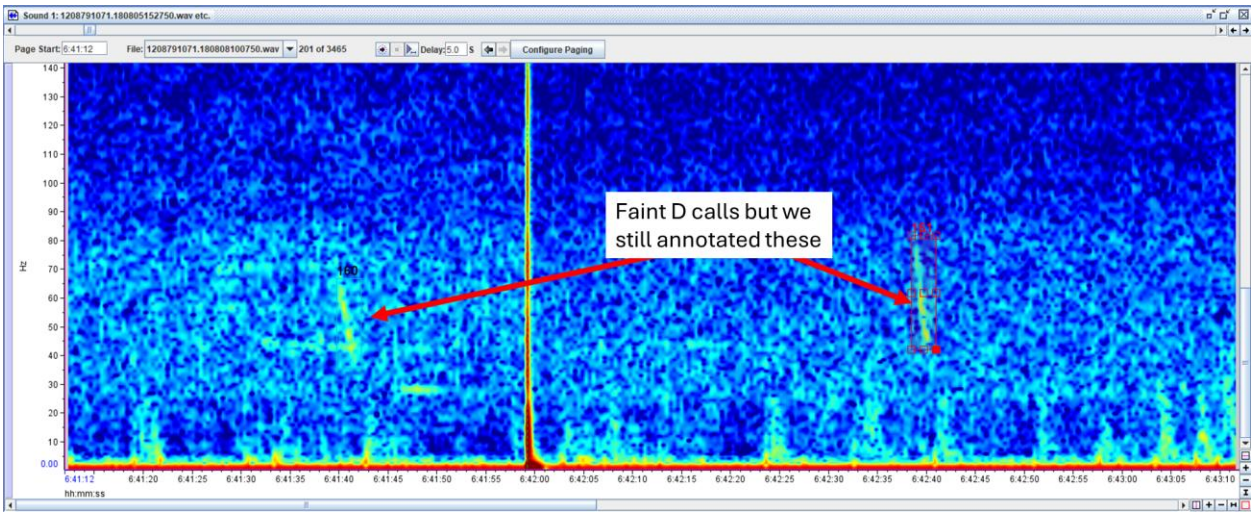



***Figure 12.*** *Faint blue whale D calls in CCES data that were still annotated.*

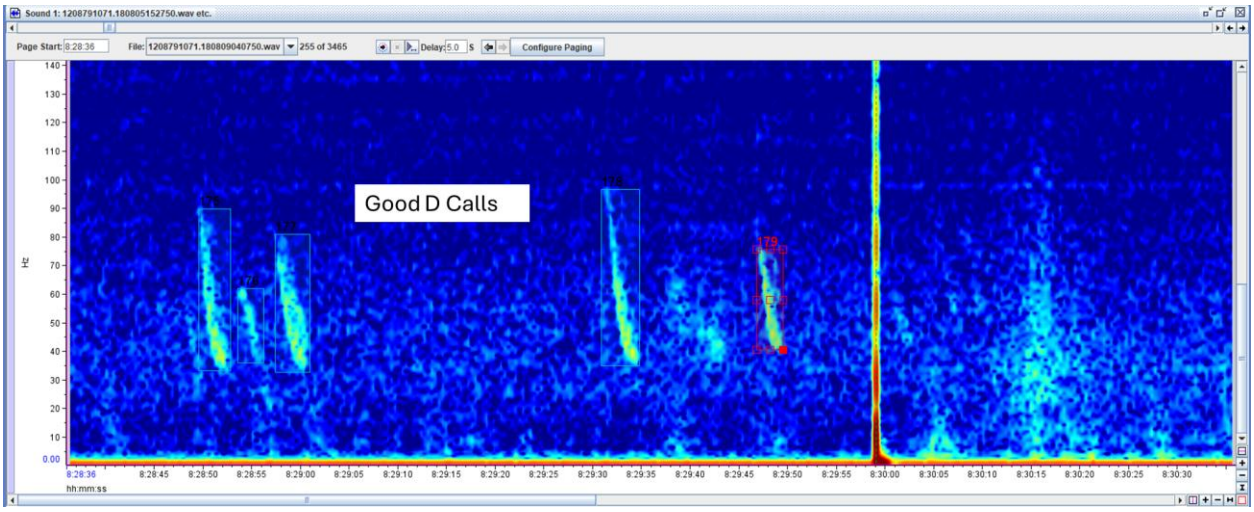



***Figure 13.*** *Clear blue whale D calls annotated in CCES data (boxes 175–179).*

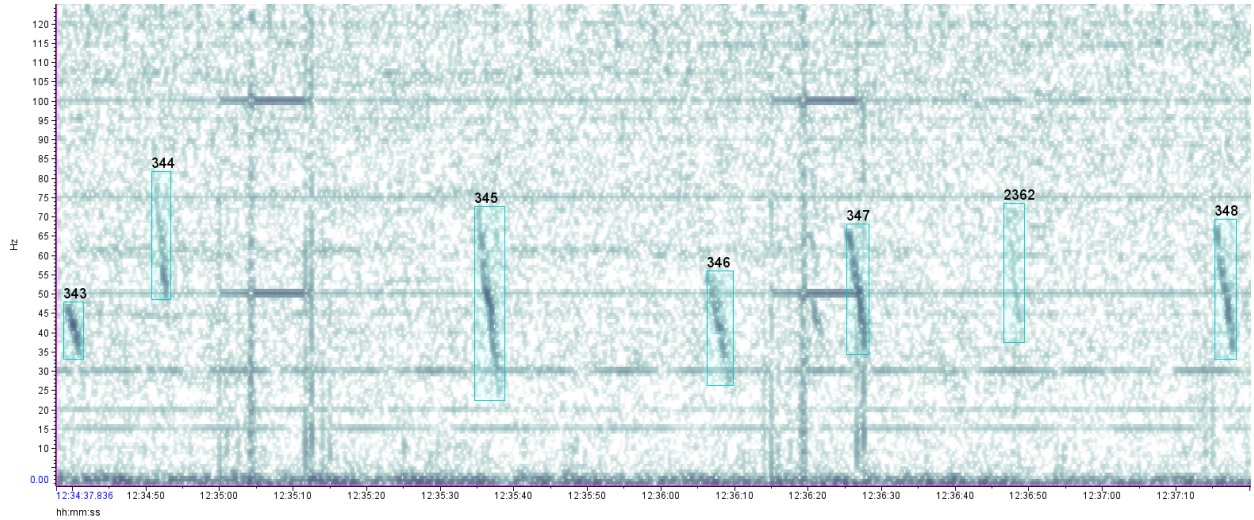



***Figure 14.*** *Blue whale D call annotations (boxes 343–348 and 2362), showing variation in the frequency range and duration of these calls.*

# SEI WHALE ANNOTATION CRITERIA:

- Relatively little is known about sei whale calls in the North Pacific. They produce downsweeps that tend to occur in series and are less variable than blue whale D calls, with an average duration of about 1–1.3 s. See Alsaid et al. (2026) for the seasonal and diel acoustic occurrence of these calls.
- Downsweeps can occur as doublets or triplets, or in longer bouts (Figures 15–20).
- Each downsweep was boxed individually, including those within doublets, triplets and bouts (Figures 21 and 22).

- In CTBTO data, the upper part of each downsweep is missing because the data are limited to 0–125 Hz (Figure 23). Fin whale calls in the CTBTO data were not annotated.

## Examples of sei whale calls:

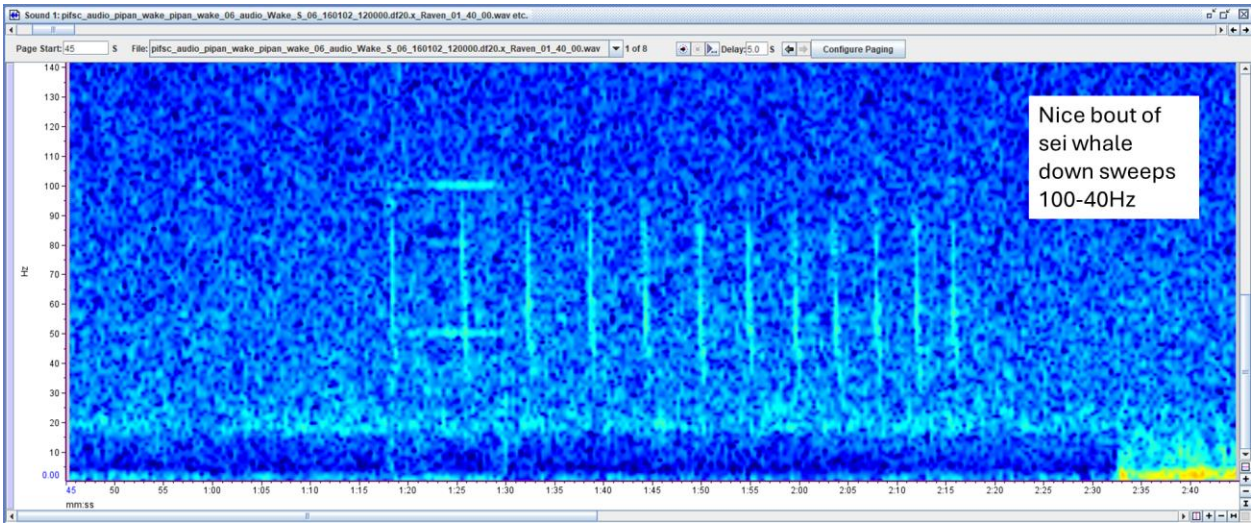



***Figure 15.*** *Bout of sei whale downsweeps sweeping from ~100 to 40 Hz in WAKE HARP data (2016).*

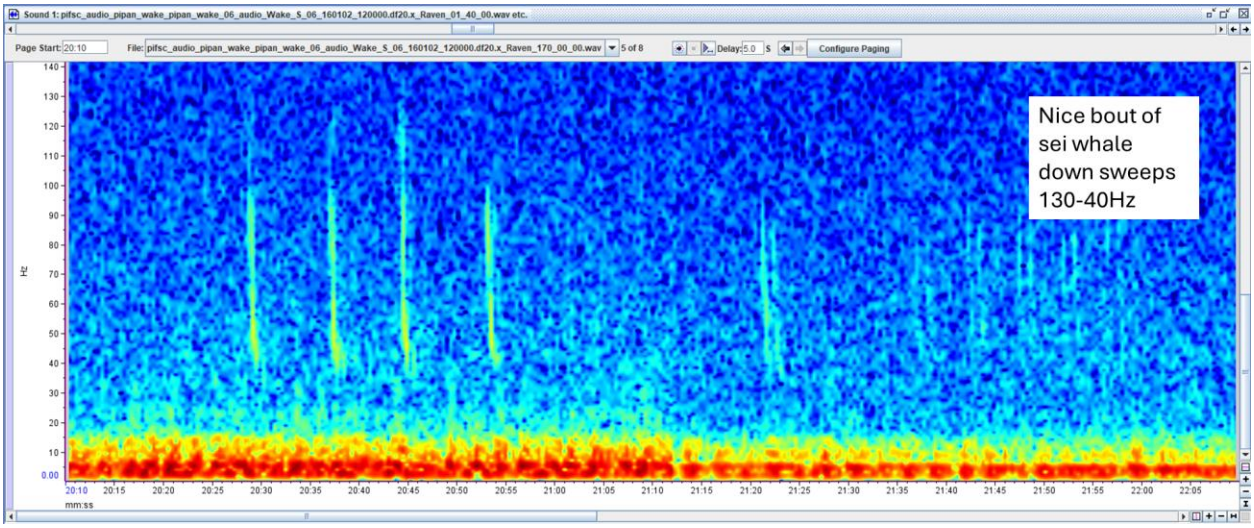



***Figure 16.*** *Bout of sei whale downsweeps sweeping from ~130 to 40 Hz in WAKE HARP data (2016).*

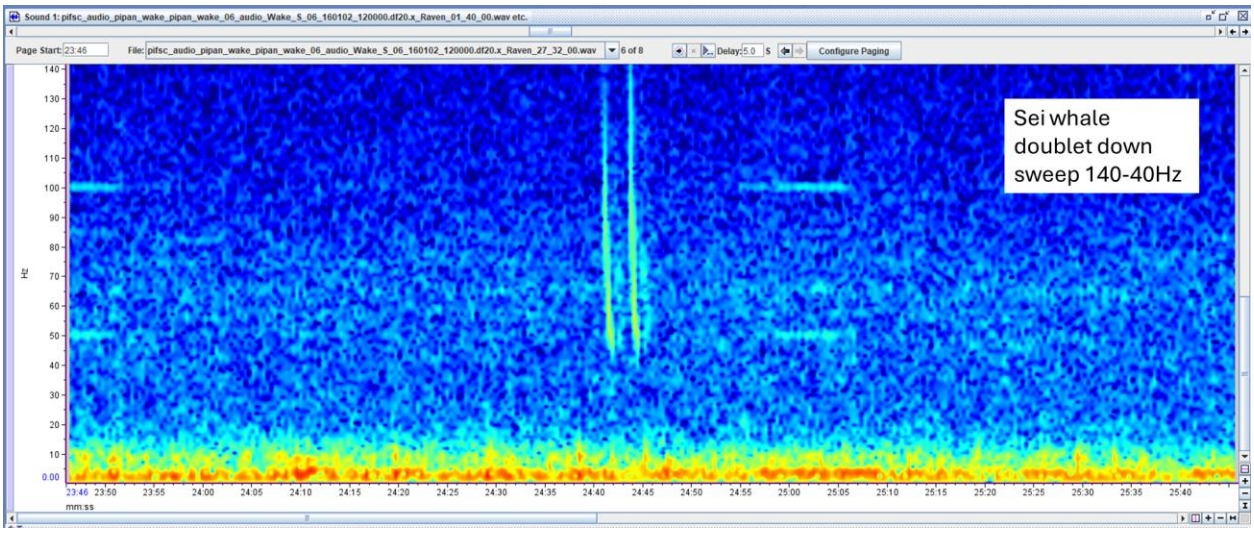



***Figure 17.*** *Sei whale downsweep doublet sweeping from ~140 to 40 Hz in WAKE HARP data (2016).*

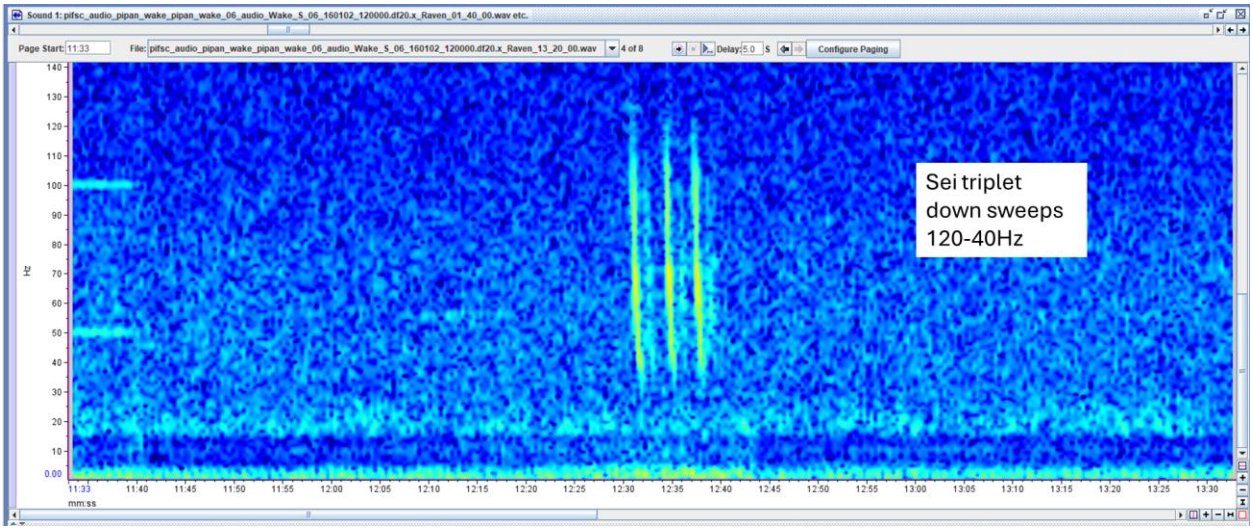



***Figure 18.*** *Sei whale downsweep triplet sweeping from ~120 to 40 Hz in WAKE HARP data (2016).*

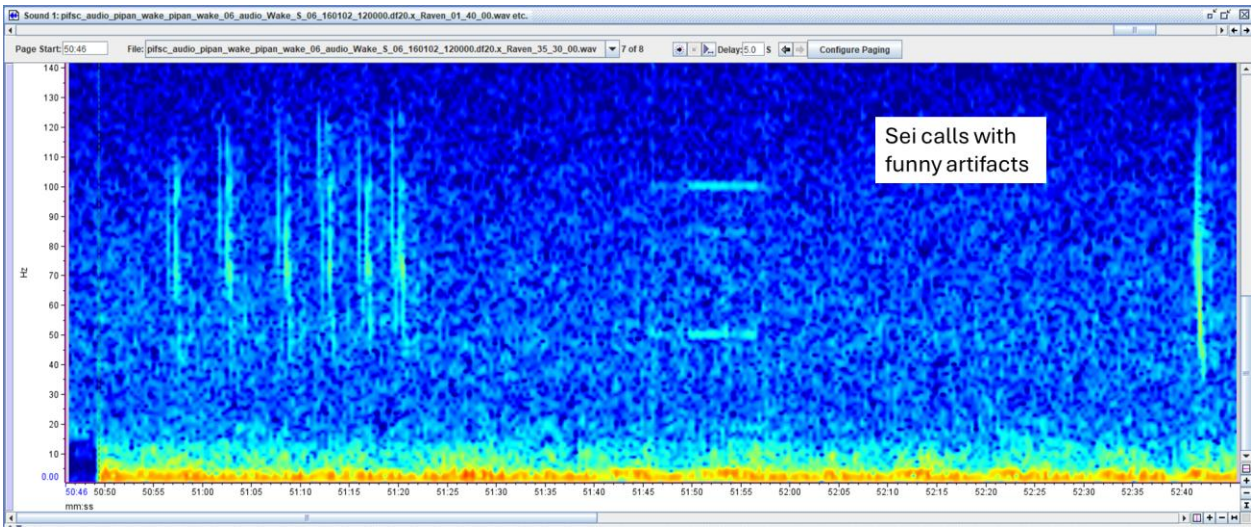



***Figure 19.*** *Sei whale downsweeps accompanied by recording artifacts in WAKE HARP data (2016).*

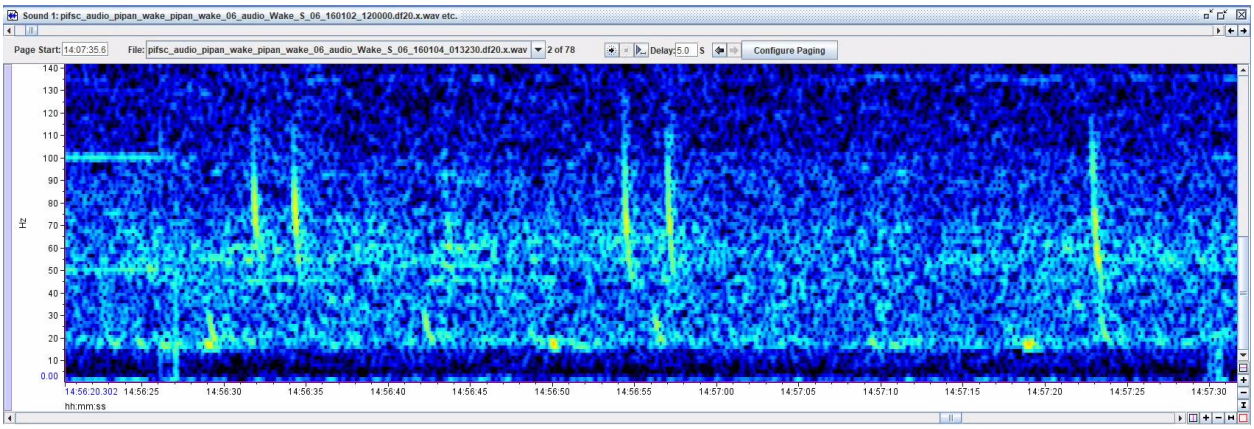


***Figure 20.*** *Sei whale downsweep doublets and a single downsweep in WAKE HARP data (2016).*

## Annotation of sei whale calls:

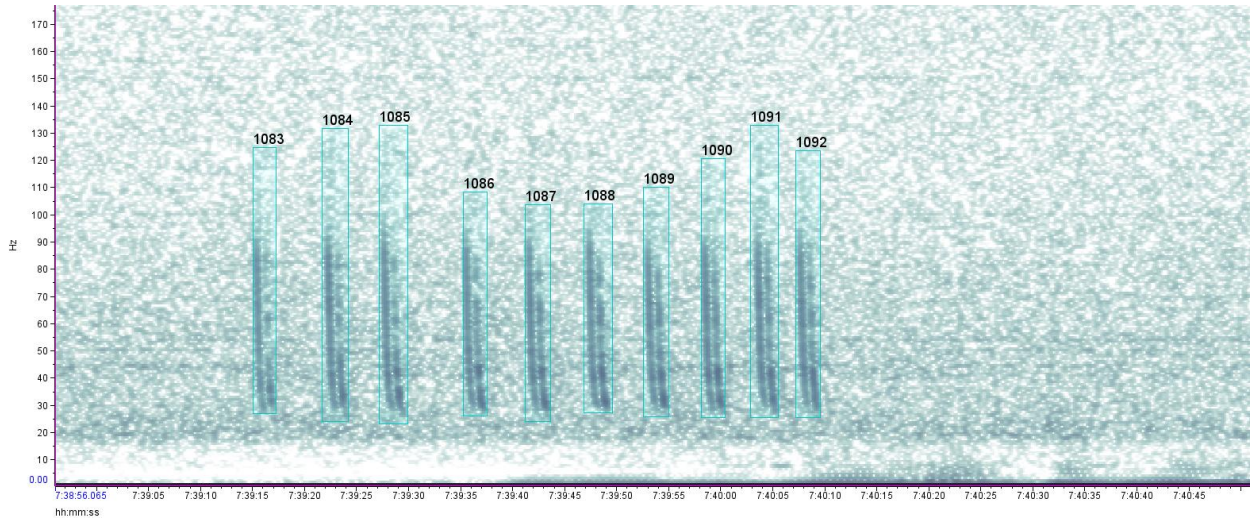



***Figure 21.*** *A series of sei whale downsweeps, each boxed individually (boxes 1083–1092).*

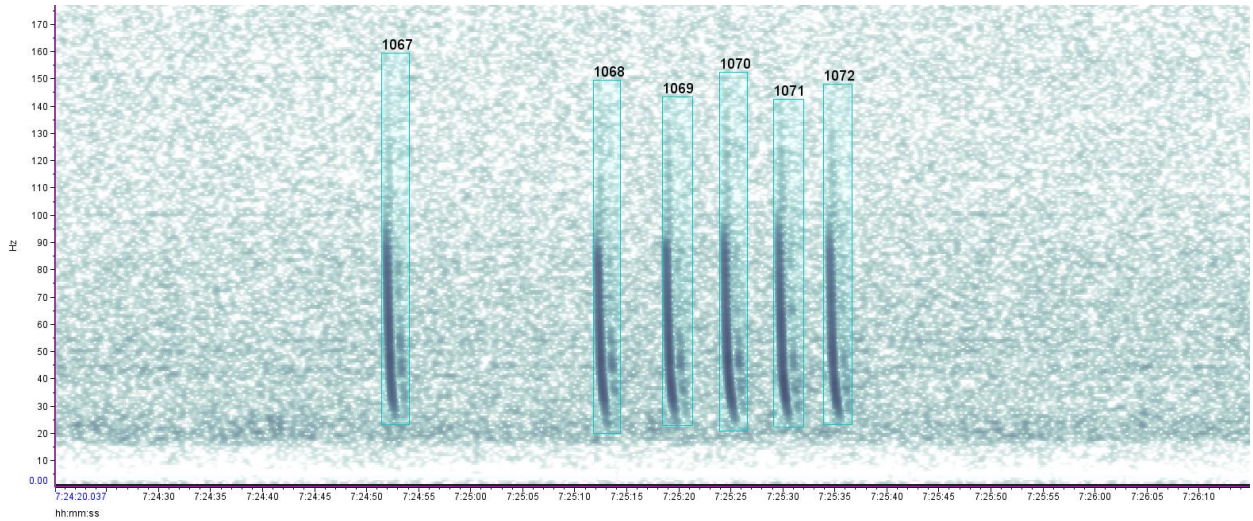



***Figure 22.*** *Sei whale downsweeps, each boxed individually (boxes 1067–1072).*

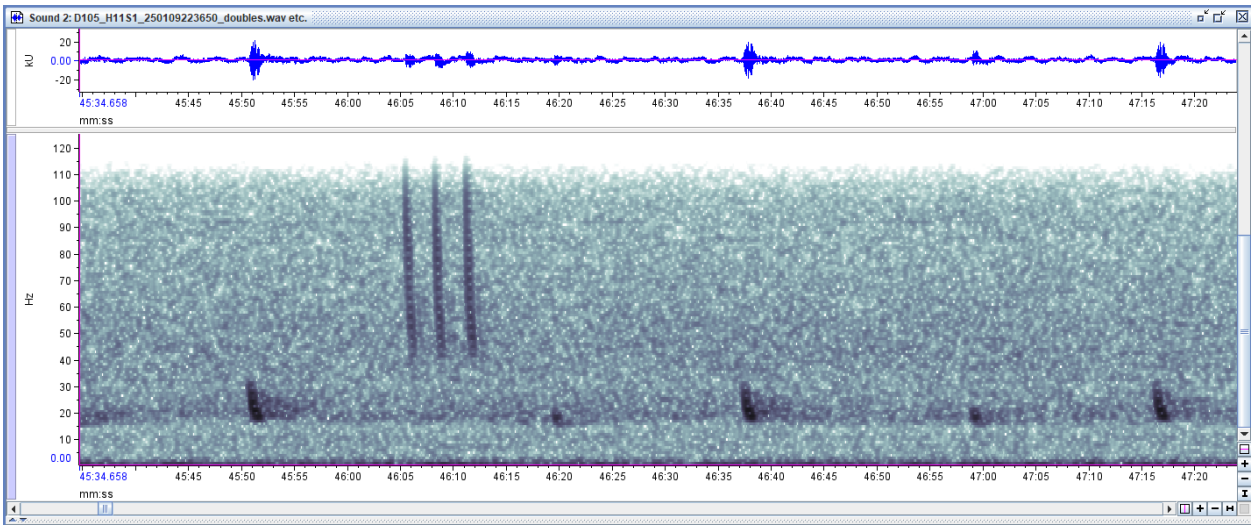



***Figure 23.*** *Waveform (top) and spectrogram (bottom) of a sei whale downsweep triplet in CTBTO HA11 data (station H11S1). The upper part of each downsweep is missing because the data are limited to 0–125 Hz. Fin whale 20-Hz pulses near 20 Hz were not annotated in this dataset.*

# MULTI-CLASS DETECTION REVIEW:

In DeepAcoustics, each detection box is labelled with its predicted call type, which makes reviewing and confirming identifications easier (Figure 24).

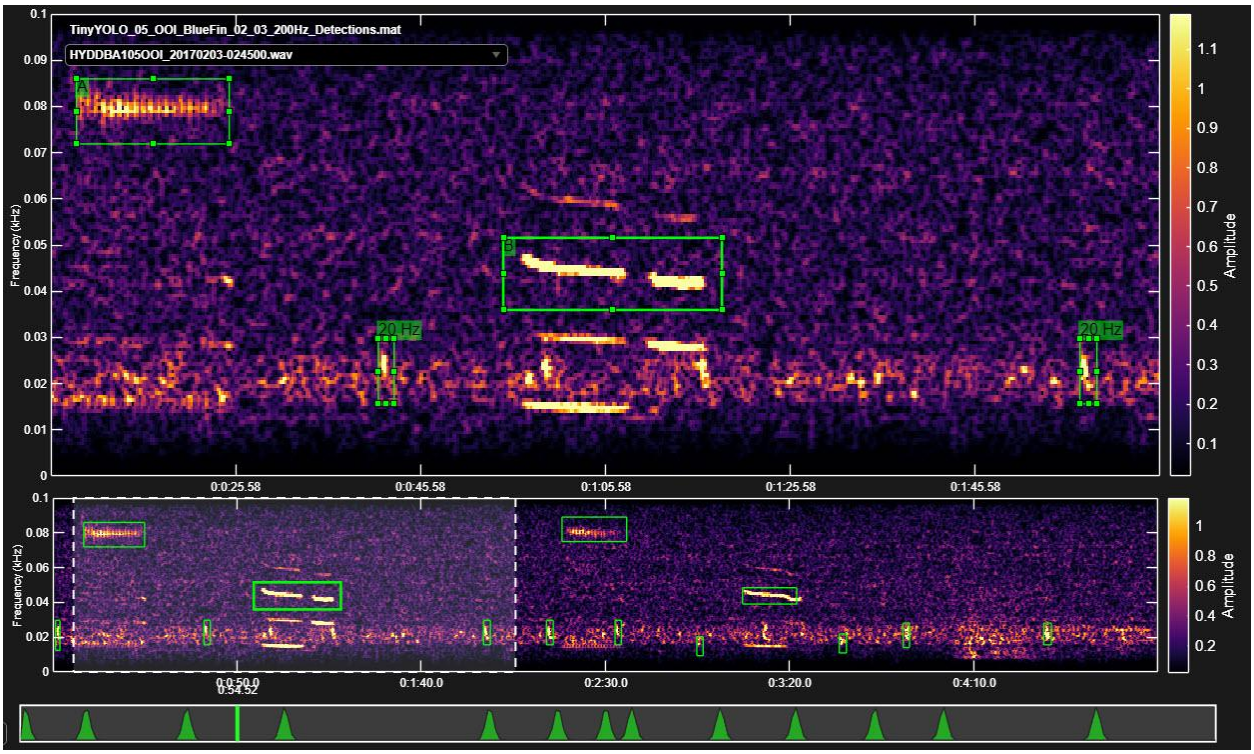



***Figure 24.*** *DeepAcoustics detection review window showing TinyYOLO detections on OOI data (HYDDBA105OOI_20170203-024500.wav). Each box is labelled with its predicted class (here blue whale A and B calls and fin whale 20-Hz pulses).*